

Department of Management Studies

Data Engineering End-To-End Cloud Project

“Azure Data Superstore Pipeline: End-to-End Data Engineering and Visualization”

Program: MBA-AI&DS
Year & Semester: III SEM

Session: 2024-2025

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Azure Data Superstore Pipeline: End-to-End Data Engineering and Visualization



databricks

Introduction

In today's data-driven world, the ability to process, analyze, and visualize large volumes of data efficiently is paramount. Organizations rely on seamless data pipelines to derive actionable insights that drive decision-making. This project is a practical implementation of a modern data engineering pipeline built using Microsoft Azure services. The goal is to process and analyze transactional sales data from the SampleSuperstore.csv dataset and present meaningful insights through interactive dashboards.

The project incorporates cutting-edge Azure technologies such as Azure Data Factory, Azure Data Lake Gen 2, Azure Synapse Analytics, and Azure Databricks, along with visualization tools like Power BI. By leveraging these tools, this project demonstrates the end-to-end process of data ingestion, transformation, storage, analysis, and visualization. It serves as a comprehensive example of how cloud-based solutions can streamline the data engineering lifecycle while ensuring scalability, reliability, and efficiency.

Project Overview

This project demonstrates the implementation of a comprehensive data engineering pipeline leveraging Microsoft Azure services. The primary objective is to process and transform the SampleSuperstore.csv dataset, which contains transactional sales data, into actionable insights. This pipeline utilizes Azure Data Factory for data movement, Azure Data Lake Gen 2 for storage, Azure Synapse Analytics for querying and analysis, Azure Databricks for data transformation, and Power BI for visualization and reporting.

Tools and Services Used

Microsoft Azure Services

1. Azure Data Factory (ADF):

- Orchestrates the movement and integration of data across various services.
- Ensures seamless automation and scheduling of data workflows.



2. Azure Data Lake Gen 2:

- Provides scalable and secure storage for structured and unstructured data.
- Serves as the repository for raw and processed datasets.



Azure Data Lake Storage Gen2

3. Azure Synapse Analytics:

- Facilitates querying and analysis of large datasets using SQL pools.
- Supports advanced analytics and data warehousing capabilities.



Azure Synapse Analytics

4. Azure Databricks:

- Enables distributed data processing and machine learning using Apache Spark.
- Performs data cleansing, transformation, and enrichment.



5. Power BI:

- Visualizes data through interactive dashboards and reports.
- Provides intuitive tools for business intelligence.



Steps in the Data Pipeline:-

Step 1: Data Collection and Storage

1. Setup Azure Data Lake Gen 2:

- Create a new storage account in Azure.

Microsoft Azure

Search resources, services, and docs (G+I)

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Home > Storage accounts >

Create a storage account

BasicsAdvancedNetworkingData protectionEncryptionTagsReview + create

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *
Azure for Students

Resource group *
(New) SuperStore_RG
[Create new](#)

Instance details

Storage account name *
superstorekartikey

Region *
(Europe) UK South
[Deploy to an Azure Extended Zone](#)

Primary service
Select a primary service

Performance *
☒ Standard: Recommended for most scenarios (general-purpose v2 account)

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Create a storage account

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[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	SuperStore_RG
Location	UK South
Storage account name	superstorekartikey
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

Security

PreviousNextCreate

superstorekartikey_1734104547760 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

InputsOutputsTemplate

✔ Your deployment is complete

Deployment name: superstorekartikey_1734104547760
Subscription: Azure for Students
Resource group: SuperStore_RG

Start time: 12/13/2024, 9:13:42 PM
Correlation ID: 57113093-537b-403d-963d-9e653605114c

Deployment details

Next steps

Go to resource

Give feedback
Tell us about your experience with deployment

Deployment succeeded

Deployment 'superstorekartikey_1734104547760' to resource group 'SuperStore_RG' was successful.

Go to resourcePin to dashboard

Cost Management

Get notified to stay within your budget and prevent unexpected charges on your bill. Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure Go to Microsoft Defender for Cloud >

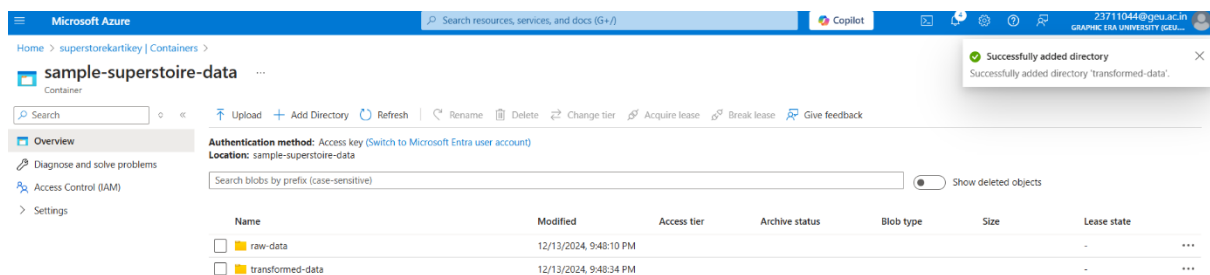
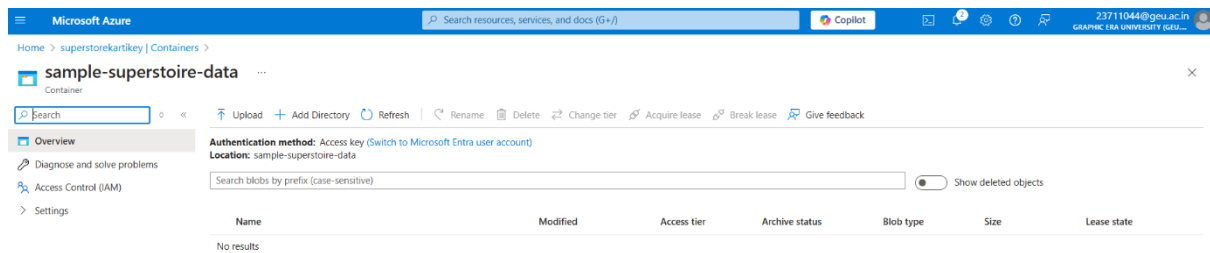
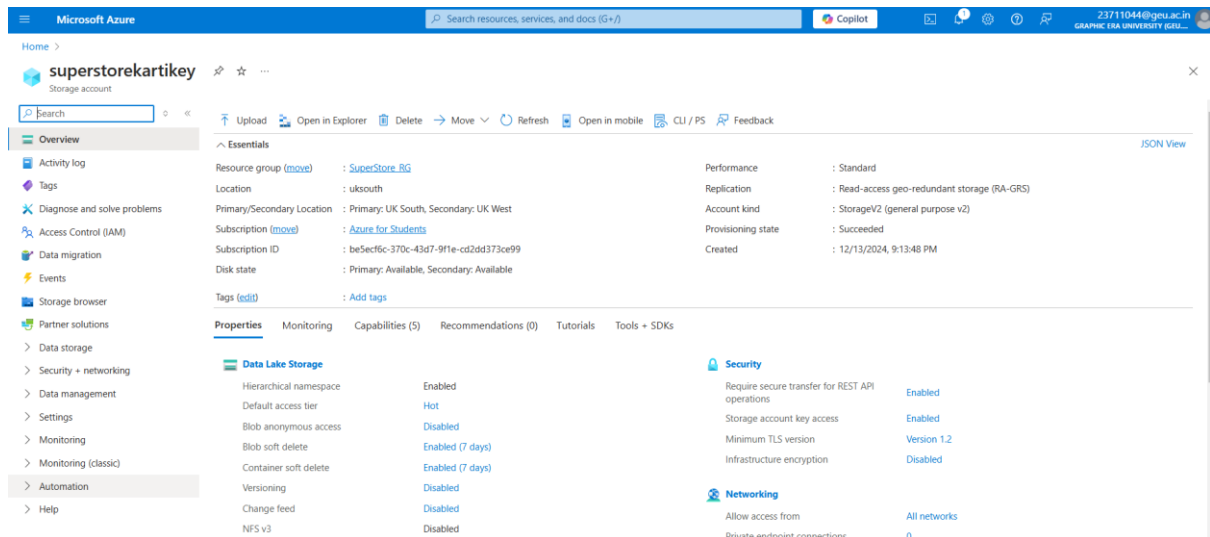
Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support. Find an Azure expert >

- Enable the hierarchical namespace to support Data Lake Gen 2 capabilities.
- Use Azure Storage Explorer to upload the SampleSuperstore.csv dataset into a designated container.



2. Dataset Description:

- The SampleSuperstore.csv dataset contains columns such as Order Date, Region, Sales, Profit, and Category, providing transactional and geographic insights.

Step 2: Data Movement Using Azure Data Factory

1. Create Azure Data Factory Instance:

- Navigate to the Azure portal and set up a new Data Factory.

Microsoft Azure

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Home > Data factories >

Create Data Factory

BasicsGit configurationNetworkingAdvancedTagsReview + create

One-click to create data factory with sample pipeline and datasets. Try it

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * SuperStore_RG

Create new

Instance details

Name * sample-superstore-df

Region * Southeast Asia

Version * V2

Microsoft Azure

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Home > Data factories >

Create Data Factory

BasicsGit configurationNetworkingAdvancedTagsReview + create

View automation template

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Basics

Subscription Azure for Students

Resource group SuperStore_RG

Name sample-superstore-df

Region Southeast Asia

Version V2

Networking

Connect via Public endpoint

PreviousNextCreate

Give feedback

Microsoft Azure

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Home >

Microsoft.DataFactory-20241213215057 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.DataFactory-20241213215057

Subscription : Azure for Students

Resource group : SuperStore_RG

Start time : 12/13/2024, 9:54:38 PM

Correlation ID : d615465c-bb44-404f-8465-66af5d398a5c

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Microsoft.DataFactory-20241213215057 > Overview

sample-superstore-df

Data factory (V2)

Search

Delete

Essentials

Resource group (move) SuperStore_RG

Status Succeeded

Location Southeast Asia

Subscription (move) Azure for Students

Subscription ID be5ec9fc-370c-43d7-9f1e-cd2d373ce99

Type Data factory (V2)

Getting started Quick start

Azure Data Factory Studio

Launch studio

Quick Starts

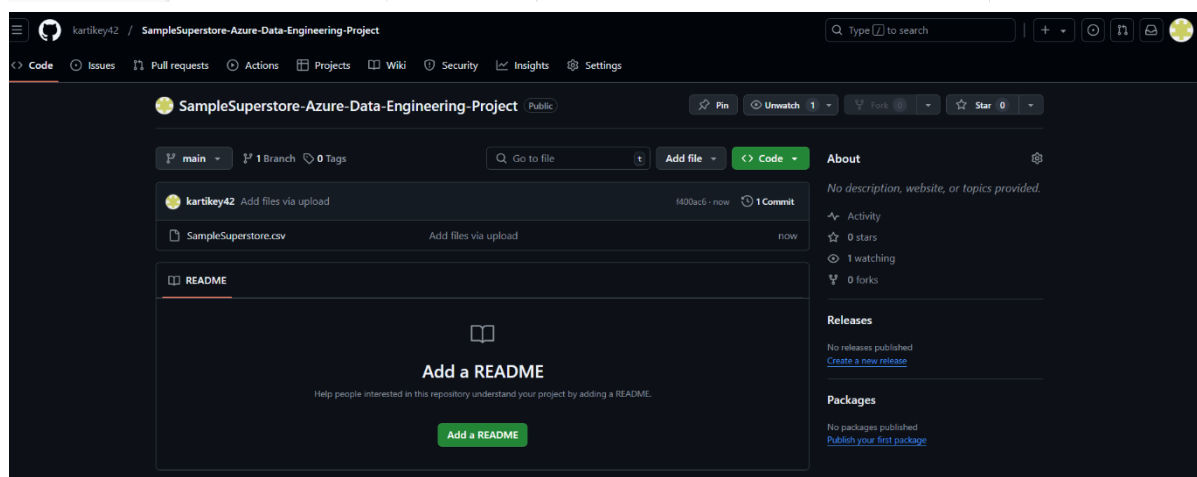
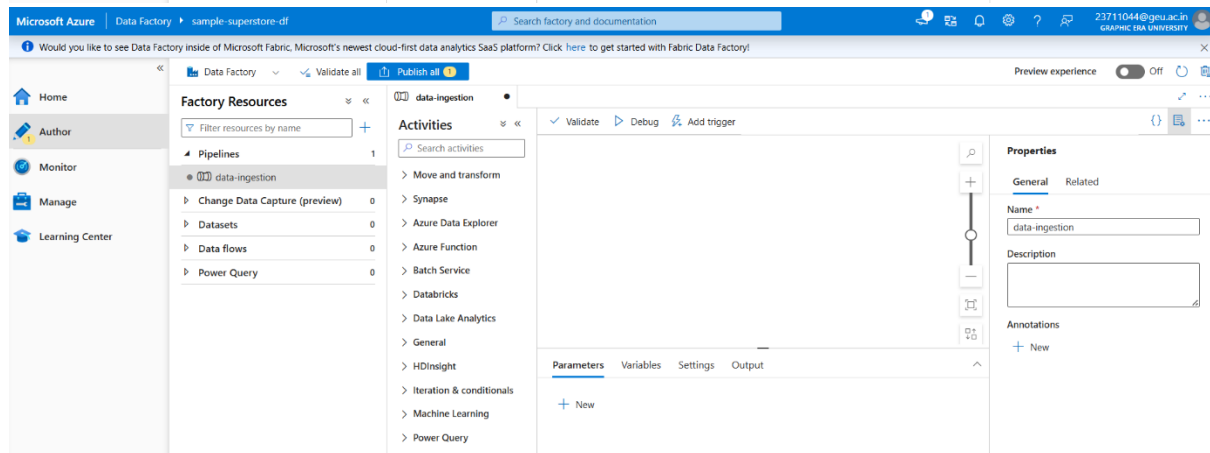
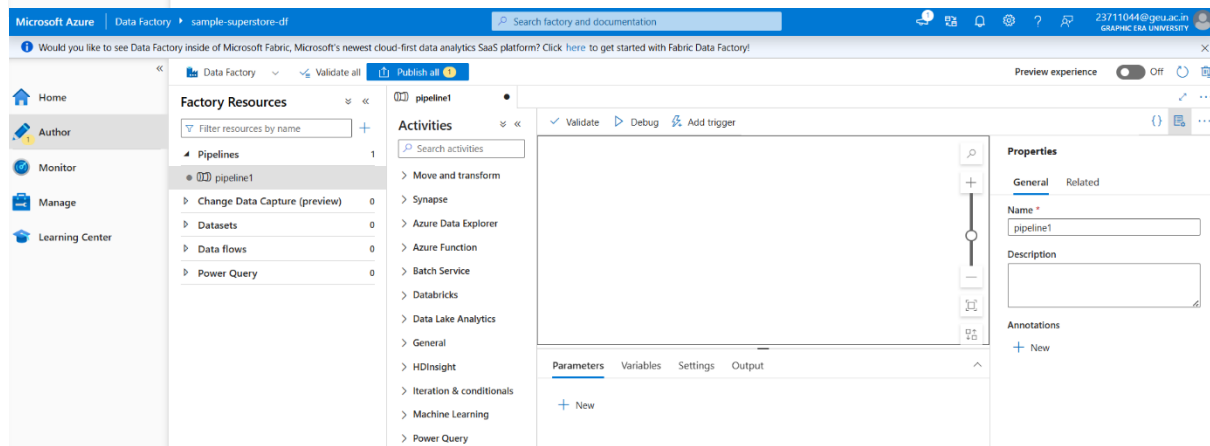
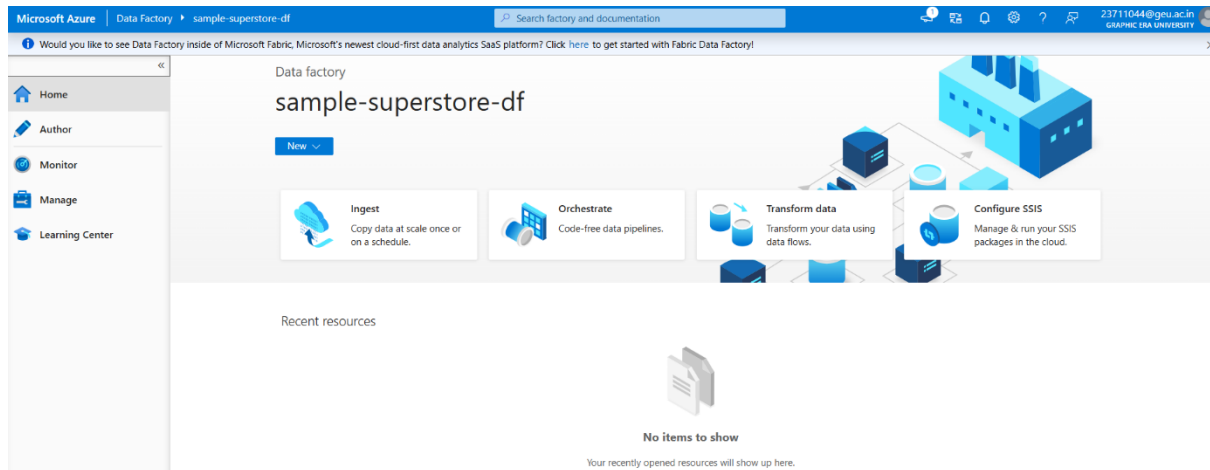
Tutorials

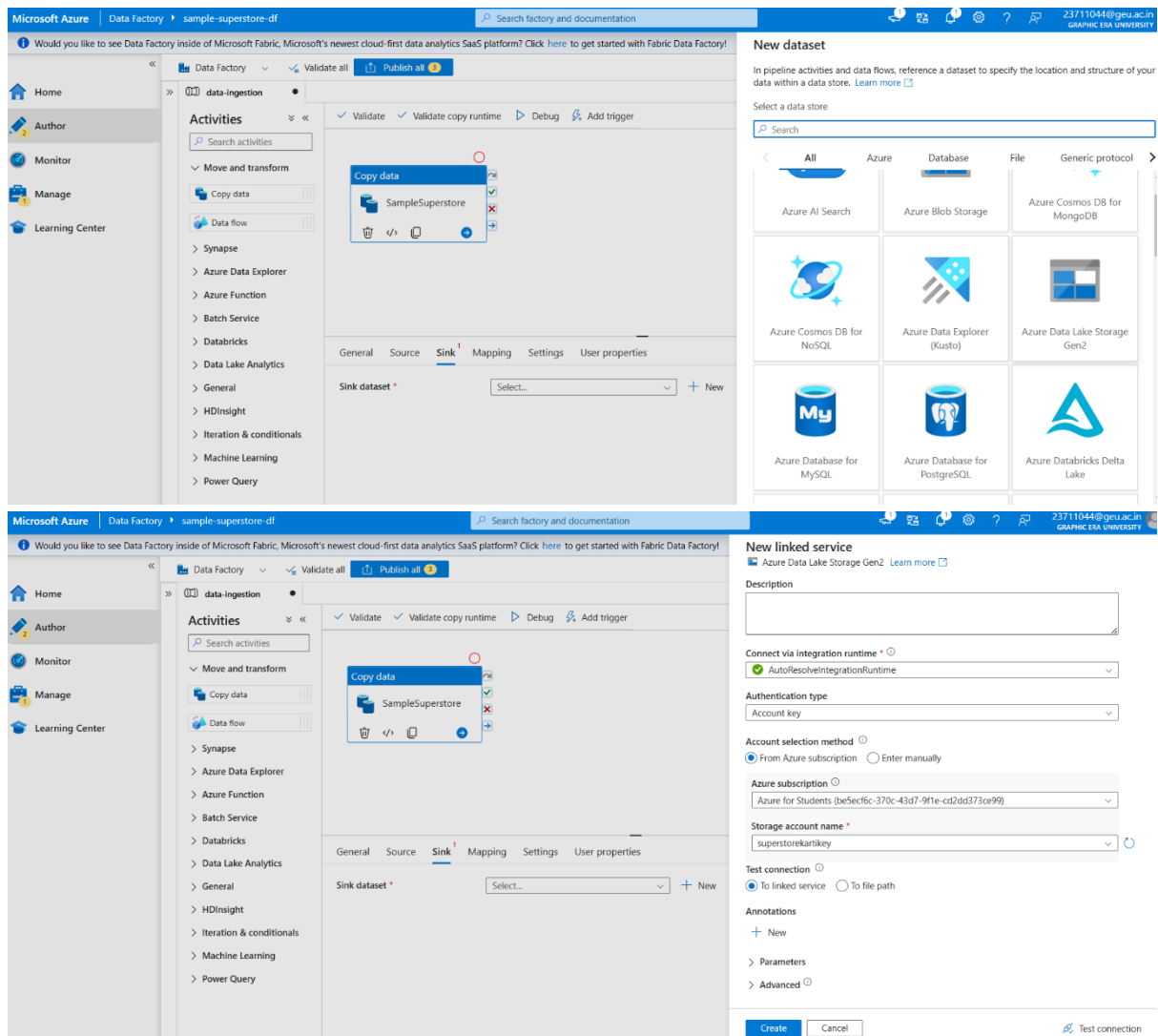
Template Gallery

Training Modules

Monitoring

- Define linked services for both the source (Data Lake) and the target (Synapse Analytics).





Set properties

Name

ADLS

Linked service *

AzureDataLakeStorage1

File path

File system

/ Directory

/ File name

First row as header



Import schema

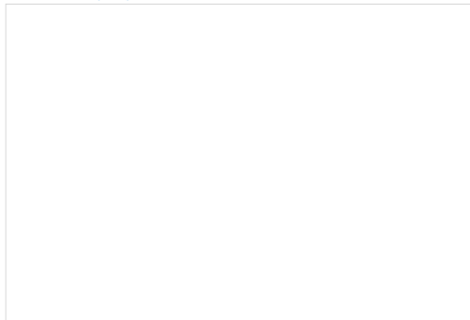
☐ From connection/store ☐ From sample file ☒ None

> Advanced

Browse

Select a file or folder.

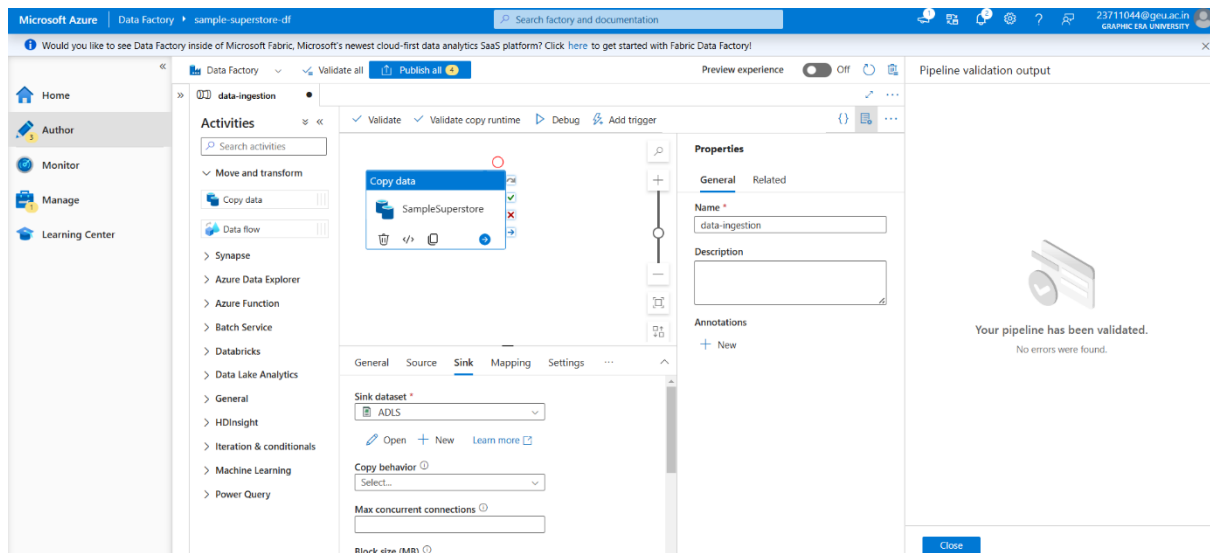
Root folder > sample-superstore-data > raw-data



Showing 0 items

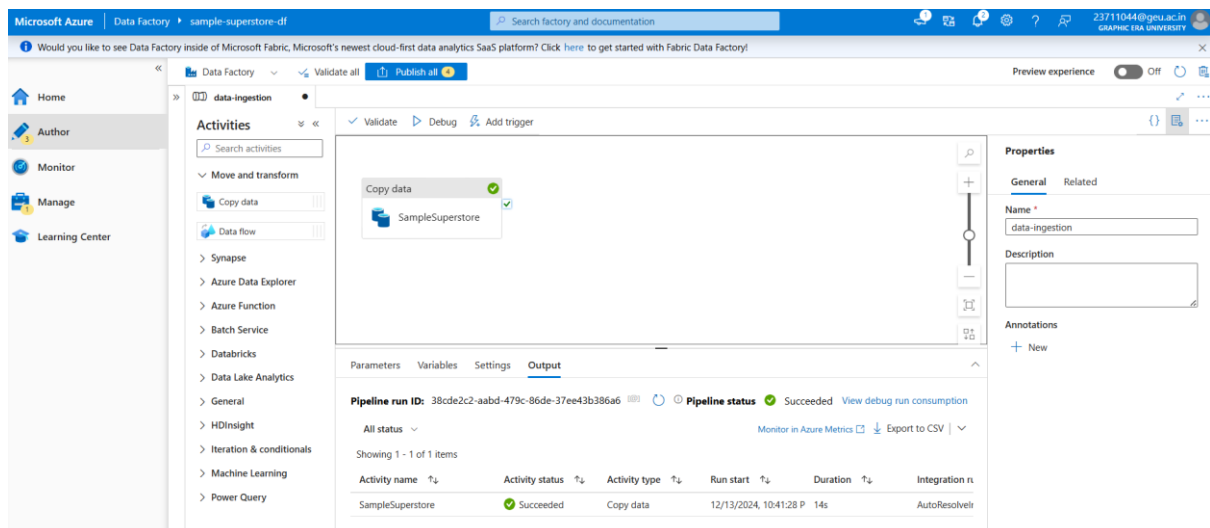
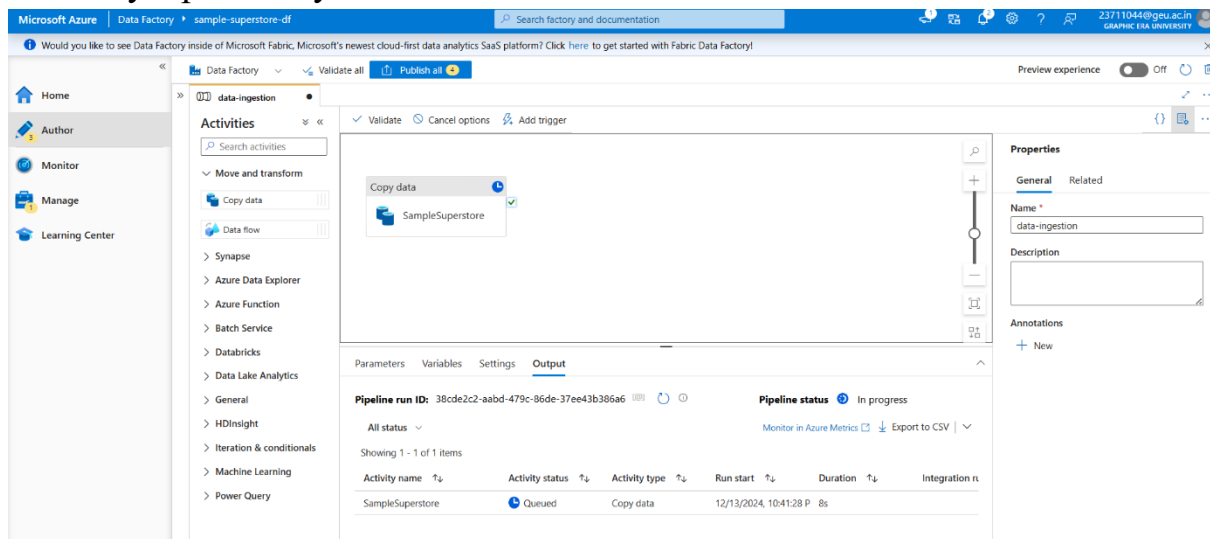
OK

Cancel



2. Pipeline Design:

- Use the **Copy Data** activity to transfer data from the Data Lake to Synapse Analytics.



- Configure additional transformations such as column mapping or filtering

The screenshot shows the Microsoft Azure portal interface. On the left, the 'sample-superstore-data' container is selected. The main pane displays the details for the 'raw-data/SampleSuperstore.csv' blob. The 'Overview' tab is active, showing properties such as URL, Last Modified, Creation Time, Version ID, Type (Block blob), Size (1.06 MiB), Access Tier (Hot), and various encryption and immutability policies.

- Automate the pipeline using scheduled triggers to ensure timely updates.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'sample-superstore-data' container is selected. The main pane displays a table of blobs. The table has columns for Name, Modified, Access tier, Archive status, Blob type, Size, and Lease state. One blob, 'SampleSuperstore.csv', is listed with a modified date of 12/13/2024, 10:41:39 PM, Hot (Inferred) access tier, and 1.06 MiB size.

Step 3: Data Transformation Using Azure Databricks

1. Setup Databricks Workspace:

- Create a Databricks workspace and configure a Spark cluster.

The first screenshot shows the 'Azure Databricks' page in the Microsoft Azure portal. It displays a message: 'No azure databricks services to display'. Below the message, there is a 'Create azure databricks service' button. The second screenshot shows the 'Create an Azure Databricks workspace' form. The form has sections for 'Project Details' and 'Instance Details'. In the 'Project Details' section, the 'Subscription' is set to 'Azure for Students' and the 'Resource group' is 'SuperStore_RG'. In the 'Instance Details' section, the 'Workspace name' is 'sample-superstore-db' and the 'Region' is 'Southeast Asia'.

Microsoft Azure

Home > Azure Databricks >

Create an Azure Databricks workspace

Validation Succeeded

Basics Networking Encryption Security & compliance Tags **Review + create**

Summary

Basics

Workspace name	sample-superstore-db
Subscription	Azure for Students
Resource group	SuperStore_RG
Region	Southeast Asia
Pricing Tier	premium
Managed Resource Group name	

Networking

Deploy Azure Databricks workspace with Secure Cluster Connectivity (No Public IP) No

Deploy Azure Databricks workspace in your own Virtual Network (VNet) No

Encryption

Enable Infrastructure Encryption No

Create < Previous Download a template for automation

Microsoft Azure

Home >

SuperStore_RG_sample-superstore-db | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview Inputs Outputs Template

Your deployment is complete

Deployment name : SuperStore_RG_sample-superstore-db
Subscription : Azure for Students
Resource group : SuperStore_RG

Start time : 12/13/2024, 10:52:57 PM
Correlation ID : d1e2f3a1-f7f8-447c-b390-2e10b7e994a8

Deployment details

Next steps

Go to resource

Give feedback
Tell us about your experience with deployment

Cost management
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- Attach the cluster to the workspace for distributed processing.

Microsoft Azure

Home > SuperStore_RG_sample-superstore-db | Overview >

sample-superstore-db

Azure Databricks Service

Search Delete

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Settings Monitoring Automation Help

Essentials

Status : Active
Resource group : SuperStore_RG
Location : Southeast Asia
Subscription : Azure for Students
Subscription ID : be5ecf6c-370c-43d7-9f1e-cd2d373ce99
Tags (edit) : Add tags

Managed Resource Group : databricks-rg-sample-superstore-db-d0rvpka7yfyg
URL : https://adb-1438387228548550.10.azure.databricks.net
Pricing Tier : Premium (+ Role-based access controls) (Click to change)

Launch Workspace

Documentation Getting Started Import Data from File Import Data from Azure Storage Notebook Admin Guide Link Azure ML workspace

Microsoft Azure databricks

sample-superstore-db

Welcome to Databricks

Search data, notebooks, recent, and more... CTRL + P

Set up your workspace
Follow this step-by-step guide that walks you through setting up the workspace for your new Databricks account.
Get started

Recents Favorites Popular Mosaic AI What's new

Start your journey
Try the "New" menu, where you can upload or connect to data and then explore it in a notebook or dashboard.
+ New

New
Workspace
Recents
Catalog
Workflows
Compute
SQL
SQL Editor
Queries
Dashboards
Genie
Alerts
Query History
SQL Warehouses
Data Engineering

Microsoft Azure databricks

Search data, notebooks, recent, and more... CTRL + P sample superstore-db

Compute

All-purpose compute Job compute SQL warehouses Vector Search Pools Policies

Filter compute you have access to Created by Only pinned Create with Personal Compute Create compute

State	Name	Policy	Runtime	Active memory	Active cores	Active DBU / h	Source	Creator	Notebooks
+ No compute Create compute to run workloads from your notebooks and jobs. Learn more about best practices for compute configuration Create compute									

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Home > App registrations

New registration Endpoints Troubleshoot Refresh Download Preview features Got feedback?

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

All applications Owned applications Deleted applications

Start typing a display name or application (client) ID to filter these results Add filters

This account isn't listed as an owner of any applications in this directory.

[View all applications in the directory](#)

Microsoft Azure Search resources, services, and docs (G+/I) Copilot 23711044@geu.ac.in GRAPHIC ERA UNIVERSITY (GEU...)

Home > App registrations > Register an application

Name

The user-facing display name for this application (this can be changed later).

app01

Supported account types

Who can use this application or access this API?

☒ Accounts in this organizational directory only (Graphic Era University only - Single tenant)
 ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant)
 ☐ Accounts in any organizational directory (Any Microsoft Entra ID tenant - Multitenant) and personal Microsoft accounts (e.g. Skype, Xbox)
 ☐ Personal Microsoft accounts only

[Help me choose...](#)

Redirect URI (optional)

We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform e.g. https://example.com/auth

By proceeding, you agree to the Microsoft Platform Policies

Register

Microsoft Azure Search resources, services, and docs (G+/I) Copilot 23711044@geu.ac.in GRAPHIC ERA UNIVERSITY (GEU...)

Home > App registrations > app01

Create application Successfully created application app01.

Search Delete Endpoints Preview features

Overview

Quickstart Integration assistant Diagnose and solve problems Manage Support + Troubleshooting

Essentials

Display name	: app01	Client credentials	: Add a certificate or secret
Application (client) ID	: 7f68c0b2-0c56-49bc-a803-68ca45455629	Redirect URIs	: Add a Redirect URI
Object ID	: 15bd3f4f-137d-4526-95dc-8a635224bc9c	Application ID URI	: Add an Application ID URI
Directory (tenant) ID	: 1490b17d-5dc9-4cbf-aeba-a2e854f521b8	Managed application in L.	: app01

Supported account types : [My organization only](#)

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

[Get Started](#) Documentation

Build your application with the Microsoft identity platform

2. Data Processing:

- Mount the Azure Data Lake Gen 2 storage to the Databricks environment for seamless data access.

Microsoft Azure portal showing the 'Certificates & secrets' page for an application registration. The 'Client secrets' tab is active, showing a table with columns for Description, Expires, Value, and Secret ID. A modal window 'Add a client secret' is open, showing a description of 'secretkey' and an expiration of 'Recommended: 180 days (6 months)'.

Application (client) ID: 7f68c8b2-0c56-49bc-a803-68ca4545629
Directory (tenant) ID : 1490b17d-5dc9-4cbf-aeba-a2e854f521b8
Secret Key: YT08Q--wT45z1f4jvR0nQO1YLh4DkdzWbCJmbJ3

```
configs = {"fs.azure.account.auth.type": "OAuth",
           "fs.azure.account.oauth.provider.type": "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
           "fs.azure.account.oauth2.client.id": "7f68c8b2-0c56-49bc-a803-68ca4545629",
           "fs.azure.account.oauth2.client.secret": "YT08Q--wT45z1f4jvR0nQO1YLh4DkdzWbCJmbJ3",
           "fs.azure.account.oauth2.client.endpoint": "https://login.microsoftonline.com/1490b17d-5dc9-4cbf-aeba-a2e854f521b8/oauth2/token"}

dbutils.fs.mount(
    source = "abfss://sample-superstore-data@superstorekartikey.dfs.core.windows.net",
    mount_point = "/mnt/samplesuperstore",
    extra_configs = configs)
```

```
configs = {"fs.azure.account.auth.type": "OAuth",
           "fs.azure.account.oauth.provider.type": "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
           "fs.azure.account.oauth2.client.id": "7f68c8b2-0c56-49bc-a803-68ca4545629",
           "fs.azure.account.oauth2.client.secret": "YT08Q--wT45z1f4jvR0nQO1YLh4DkdzWbCJmbJ3",
           "fs.azure.account.oauth2.client.endpoint": "https://login.microsoftonline.com/1490b17d-5dc9-4cbf-aeba-a2e854f521b8/oauth2/token"}

dbutils.fs.mount(
    source = "abfss://sample-superstore-data@superstorekartikey.dfs.core.windows.net",
    mount_point = "/mnt/samplesuperstore",
    extra_configs = configs)

True
```


Microsoft Azure | Search resources, services, and docs (G+I) | Copilot | 23711044@geu.ac.in

Home > Storage accounts > superstorekartikey | Containers > sample-superstore-data | Access Control (IAM)

Add role assignment

Role | Members | Conditions | Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles | Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Search by role name, description, permission, or ID | Type: All | Category: All

Name	Description	Type	Category	Details
Reader	View all resources, but does not allow you to make any changes.	BuiltInRole	General	View
App Compliance Automation Administrator	Create, read, download, modify and delete reports objects and related other resource objects.	BuiltInRole	None	View
App Compliance Automation Reader	Read, download the reports objects and related other resource objects.	BuiltInRole	None	View
Avere Contributor	Can create and manage an Avere vFXT cluster.	BuiltInRole	Storage	View

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Home > Storage accounts > superstorekartikey | Containers > sample-superstore-data | Access Control (IAM)

Add role assignment

Role | **Members** | Conditions | Review + assign

Selected role | Storage Blob Data Contributor

Assign access to | ☒ User, group, or service principal | ☐ Managed identity

Members | + Select members

Name	Object ID	Type
app01	1445219d-ac3c-476f-b7ad-04858deb42...	App
app01	46990c37-f83e-4b4b-9058-e45d6612bf...	App
app01	7c30b155-3a8d-40d2-a056-479677cd78...	App

Description | Optional

[Review + assign](#) | [Previous](#) | [Next](#) | [Feedback](#)

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Home > Storage accounts > superstorekartikey | Containers > sample-superstore-data

sample-superstore-data | Access Control (IAM)

Overview | **Check access** | Role assignments | Roles | Deny assignments | Classic administrators

Diagnose and solve problems | **Access Control (IAM)** | Settings

My access | View my level of access to this resource. | [View my access](#)

[Add](#) | [Download role assignments](#) | [Edit columns](#) | [Refresh](#) | [Delete](#) | [Feedback](#)

Added Role assignments | 3 Role assignments were added as Storage Blob Data Contributor for sample-superstore-data.

- Use Spark DataFrame APIs for reading, transforming, and writing data.

Microsoft Azure | databricks | Search data, notebooks, repects, and more... | CTRL + P | sample-superstore-db

New | Workspace | Recents | Catalog | Workflows | Compute | SQL | SQL Editor | Queries | Dashboards | Genie | Alerts | Query History | SQL Warehouses | Data Engineering | Job Runs | Data Ingestion | Delta Live Tables | Machine Learning

SampleSuperstore Data Transformation

File | Edit | View | Run | Help | Last edit was 6 minutes ago | [Run all](#) | [Schedule](#) | [Share](#)

```

1
dbutils.fs.mount(
  source = "abfss://sample-superstore-data@superstorekartikey.dfs.core.windows.net",
  mount_point = "/mnt/samplestore",
  extra_configs = configs)

True

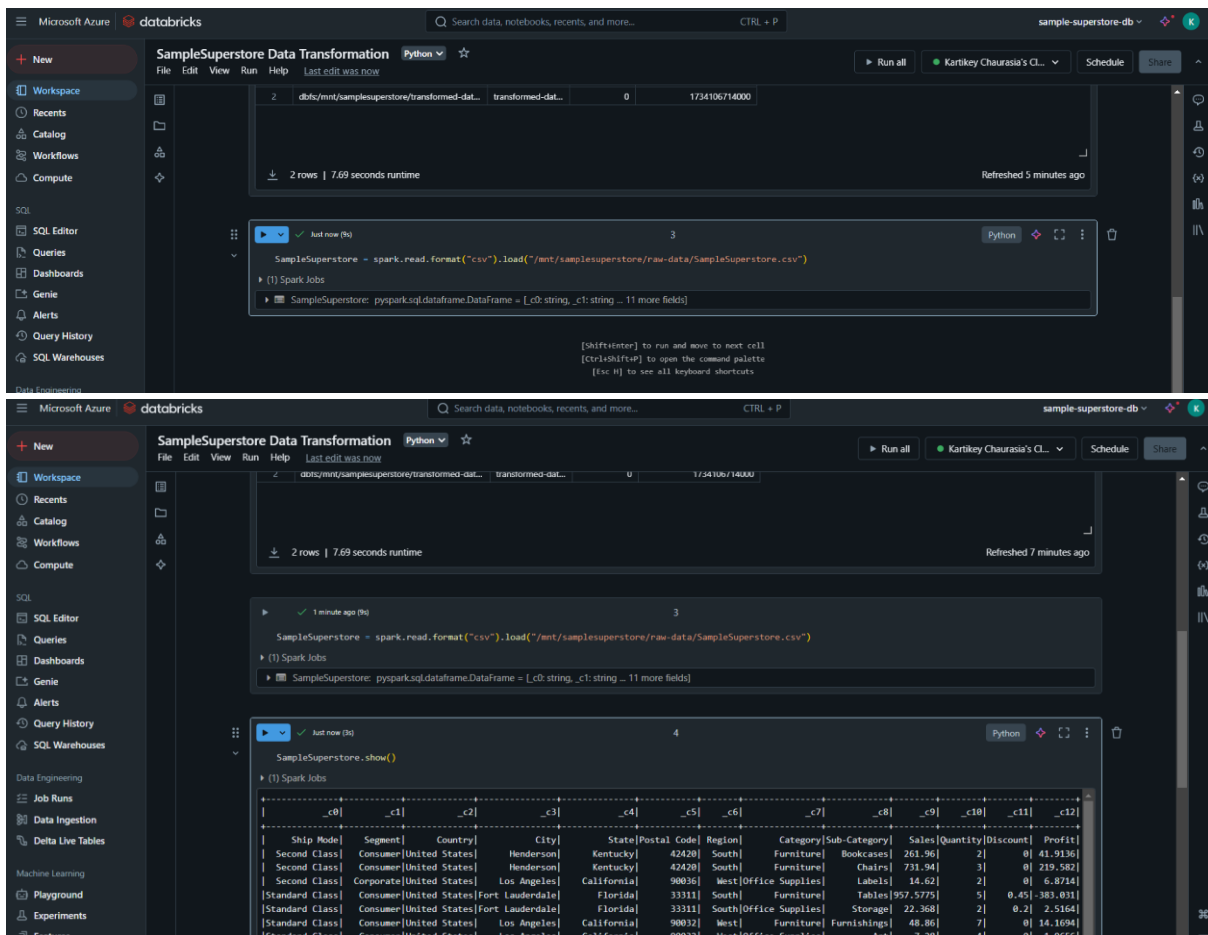
```

2 | Just now (1s) | Python | [Refresh](#)

%fs
ls /mnt/samplestore

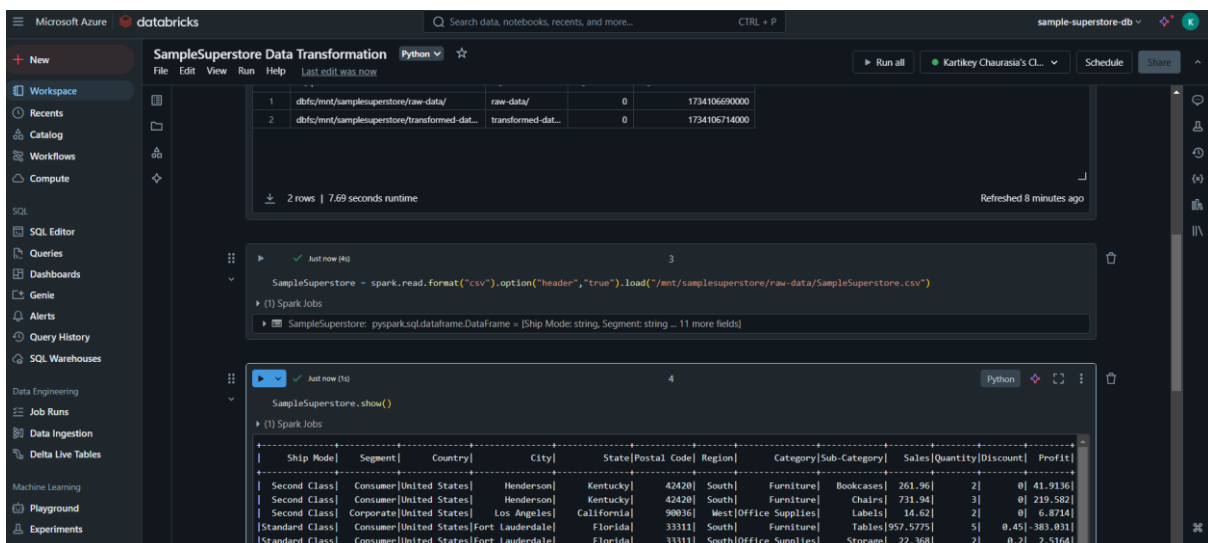
path	name	size	modificationTime
dbfs/mnt/samplestore/raw-data/	raw-data/	0	1734106900000
dbfs/mnt/samplestore/transformed-dat...	transformed-dat...	0	1734106714000

2 rows | 7.69 seconds runtime | Refreshed now



3. Transformation Activities:

- Handle missing data by filling or removing null values.
- Remove duplicate rows to ensure data integrity.
- Calculate derived metrics, such as Profit Margin using formula-based transformations.



4. Write Processed Data:

- Save the transformed dataset back to Data Lake Gen 2 or directly into Synapse Analytics for further analysis.

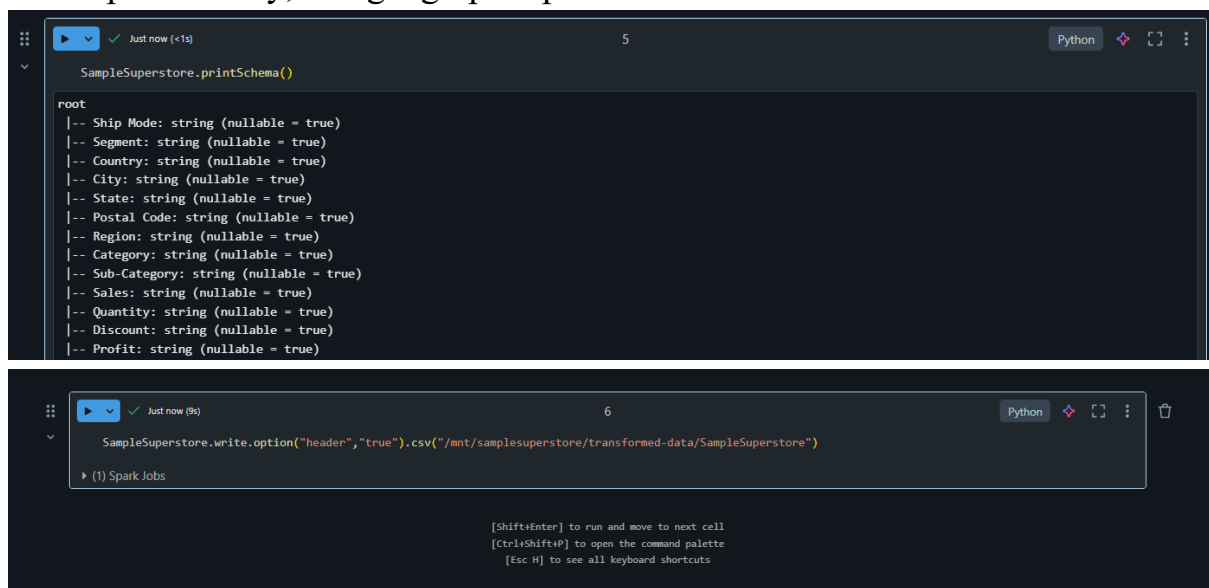
Step 4: Data Analysis Using Azure Synapse Analytics

1. Setup Synapse SQL Pools:

- Configure Synapse Analytics SQL pools to optimize data storage and retrieval.
- Load the transformed dataset into Synapse tables.

2. SQL Queries:

- Perform detailed analysis using SQL queries to identify sales trends, profitability, and geographic performance.



The image displays two screenshots of a Jupyter Notebook environment. The top screenshot, labeled '5', shows a code cell with the command `SampleSuperstore.printSchema()`. The output displays a schema for the `SampleSuperstore` dataset, listing various fields such as `Ship Mode`, `Segment`, `Country`, `City`, `State`, `Postal Code`, `Region`, `Category`, `Sub-Category`, `Sales`, `Quantity`, `Discount`, and `Profit`, each with a nullable status. The bottom screenshot, labeled '6', shows a code cell with the command `SampleSuperstore.write.option("header", "true").csv("/mnt/samplesuperstore/transformed-data/SampleSuperstore")`. Below the code cell, there is a section for Spark Jobs and a list of keyboard shortcuts: `[Shift+Enter]` to run and move to next cell, `[Ctrl+Shift+P]` to open the command palette, and `[Esc H]` to see all keyboard shortcuts.

3. Performance Optimization:

- Create indexes and materialized views to enhance query execution speed.

Step 5: Visualization Using Power BI

1. Connect Power BI to Synapse Analytics:

- Use the native Azure Synapse connector in Power BI Desktop to access processed data.

2. Dashboard Creation:

- Design dashboards featuring:
 - Total and regional sales metrics.
 - Profitability trends across product categories.
 - Comparative analysis by customer segments and regions.

3. Sharing Insights:

- Publish the dashboard to the Power BI Service.
- Configure access controls and sharing permissions for stakeholders.

Power BI Dashboard

Sample Superstore Profit Report

- State
- ☐ Alabama
- ☐ Arizona
- ☐ Arkansas
- ☐ California
- ☐ Colorado
- ☐ Connecticut
- ☐ Delaware
- ☐ District of Columbia
- ☐ Florida
- ☐ Georgia
- ☐ Idaho
- ☐ Illinois
- ☐ Indiana
- ☐ Iowa
- ☐ Kansas
- ☐ Kentucky
- ☐ Louisiana
- ☐ Maine
- ☐ Maryland
- ☐ Massachusetts

2.30M 286....

Sum of Sales

Sum of Profit

38K

Sum of Quantity

Sum of Profit

by Year and Quarter



Sum of Sales by Region



Sum of Profit

by Region



Sum of Profit

by Segment



Sum of Profit

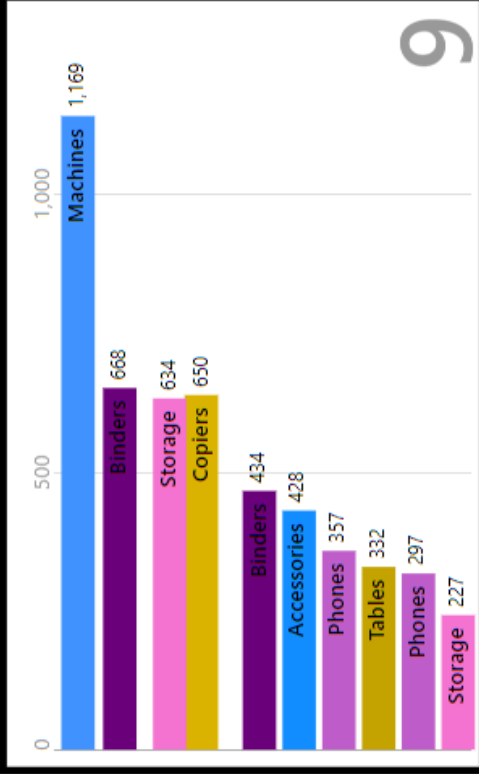
by Category



Sum of Quantity	Sub-Category	Sum of Profit
2976	Accessories	41,900
1729	Appliances	18,100
3000	Art	6,500
5974	Binders	30,200
868	Bookcases	-3,400
2356	Chairs	26,500
234	Copiers	55,600
906	Envelopes	6,900
914	Fasteners	9,000
3563	Furnishings	13,000
1400	Labels	5,500
440	Machines	3,300
5178	Paper	34,000
3289	Phones	44,500
3158	Storage	21,200
647	Supplies	-1,100
1241	Tables	-17,700
37873		2,86,300

Sum of Profit

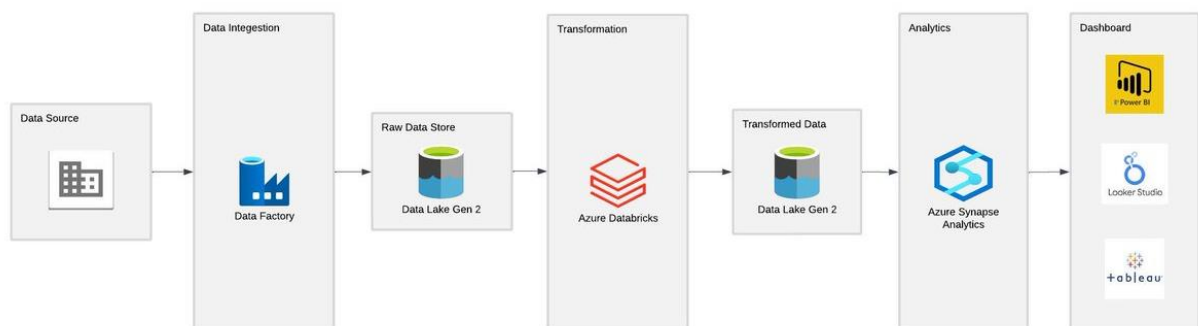
by Sub-Category, Year, Quarter, Month and Day



Project Architecture Diagram

The architecture of the project includes the following components:

- ❖ **Data Lake Gen 2:** Serves as the centralized storage for raw and transformed datasets.
- ❖ **Azure Data Factory:** Orchestrates data transfer and ensures workflows are automated.
- ❖ **Azure Databricks:** Performs distributed processing and complex data transformations.
- ❖ **Synapse Analytics:** Stores processed data and provides analytical capabilities.
- ❖ **Power BI:** Visualizes data and provides actionable insights to stakeholders.



Challenges Faced and Solutions

1. Large Dataset Handling:

- **Challenge:** Initial slow query performance in Synapse Analytics.
- **Solution:** Implemented data partitioning and indexing strategies to enhance performance.

2. Access Issues:

- **Challenge:** Difficulty in connecting Azure services due to insufficient permissions.
- **Solution:** Configured role-based access control (RBAC) and granted

appropriate permissions.

3. Data Transformation Performance:

- **Challenge:** Slow processing of large datasets in Databricks.
- **Solution:** Utilized optimized Spark configurations and parallel processing.

Conclusion

This project highlights the integration of Azure cloud technologies to create an end-to-end data engineering solution. It ensures efficient data processing, robust analysis, and insightful visualization, offering scalable and reliable tools for modern business intelligence needs.

Appendix

Prerequisites

- Active Azure Subscription.
- Power BI Desktop installed on a local machine.
- Familiarity with SQL, Python, and Azure cloud services.